

Stereoscopic Transparent Display Visible with Naked Eye

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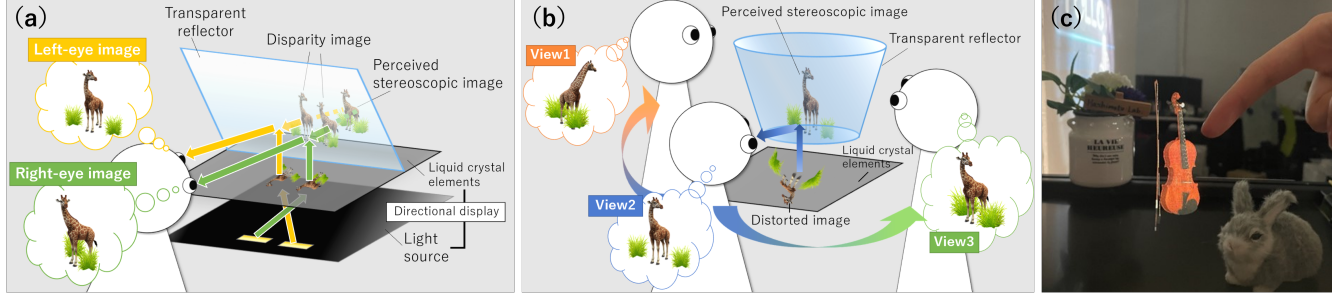


Figure 1: Overview of proposed method. (a) Principle of presenting stereoscopic images to naked eye on transparent reflector. (b) Images seen from various viewpoints. (c) Image seen by observer (right-eye viewpoint).

ABSTRACT

In this study, we propose a stereoscopic transparent display that can be viewed with the naked eye. Existing methods for generating realistic high-quality stereoscopic images require wearable devices and the presence of a display, which degrades the sense of presence. Our method increases the sense of presence by making the stereoscopic images blend into the surrounding environment.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Auto Stereoscopy, Transparent Display, Stereoscopic Display, 3D

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1 INTRODUCTION

Various stereoscopic image presentation technologies, which give images depth, have been proposed. To present a stereoscopic image that cannot be distinguished from reality, a suitable viewing environment is required. That is, the visibility of the stereoscopic display itself should be minimized and the image should be visible with the naked eye.

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Current stereoscopic displays reduce the sense of presence. Transparent stereoscopic displays have thus been proposed [Luo et al. 2017; Nayar and Anand 2006]. However, these displays have low resolution or cannot generate images that can be seen with the naked eye. A method that uses retro-transmissive optics to present stereoscopic images to the naked eye has been proposed [Kurogi et al. 2018], but observation is limited to a small number of viewpoints.

2 PROPOSED APPROACH

In this research, we propose a directional display that presents separate images to the left and right eyes to generate stereoscopic images that can be viewed with the naked eye. The stereoscopic images are presented inside a transparent reflector using Pepper's ghost technique, an illusion technique for presenting virtual images, and can be observed from various viewpoints. An overview of the method is shown in Figure 1(a).

In the proposed method, the light source and liquid crystal elements of a typical liquid crystal display are separated and placed at a distance from each other, as shown in Figure 2. This decreases the angle of incidence from the light source to the image presentation part (i.e., liquid crystal elements), thereby narrowing the diffuse (viewing) area of the light passing through the liquid crystal elements. Stereoscopic images are generated using this directional display by adjusting the display position of the disparity images for the right and left eyes and the position of the light source according to the observer's position, and switching between the disparity images and the light source at high speed.

A transparent reflector with a conical trapezoidal shape [Luo et al. 2017], shown in Figure 1(b), is used to separate incident light into reflected and transmitted components. By applying geometric correction to the image displayed on the directional display in advance and using Pepper's ghost technique, distortion-free disparity images are presented inside the transparent reflector, allowing the observer to perceive a stereoscopic image. By changing the display position of the image according to the position of the observer, the

stereoscopic image can be observed from all sides. Unlike methods that use optical elements such as retro-transmissive optics, our method does not degrade the resolution of the directional display (i.e., the resolution of the presented stereoscopic images is the resolution of the display).

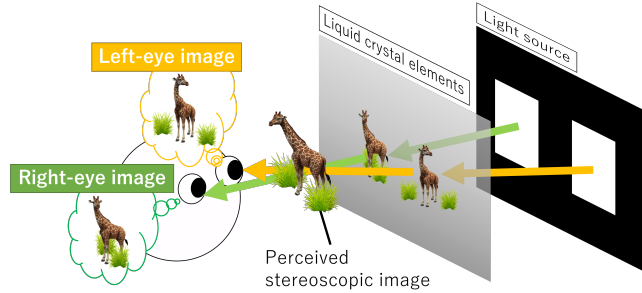


Figure 2: Principle of proposed directional display.

3 RESULTS

A prototype of the proposed method is shown in Figure 3. The arrangement of the components and the position of the presented images are shown in Figure 4. In this implementation, a 43-inch display with a brightness of 750cd/m^2 was used as the light source for the directional display. For the image presentation part, the liquid crystal elements from a 27-inch display (from which the backlight was removed) with a resolution of 1920×1080 were used. A flat polyethylene terephthalate sheet was used as the transparent reflector.

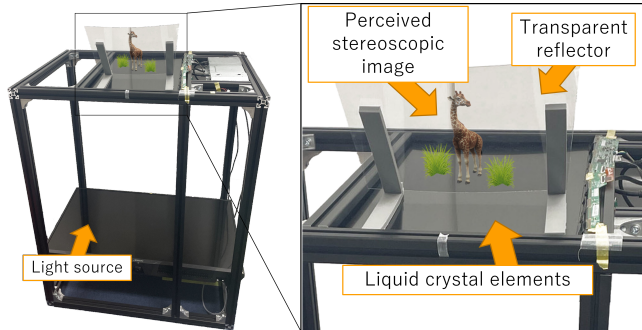


Figure 3: Photograph of prototype display (front view).

In an experiment, the prototype presented a three-dimensional model at multiple depth positions. The left- and right-eye disparity images were captured from the observer's left- and right-eye positions, respectively. The disparity images and their luminance values were measured through the transparent reflector. The obtained disparity images and the average luminance values for five measurements are shown in Figure 5(a). As shown, the disparity images can be clearly observed at the target left- and right-eye positions. The luminance value for the disparity image at the target eye is at least 25 times higher than that at the opposite eye, indicating

that the prototype system has sufficient directionality to present stereoscopic images.

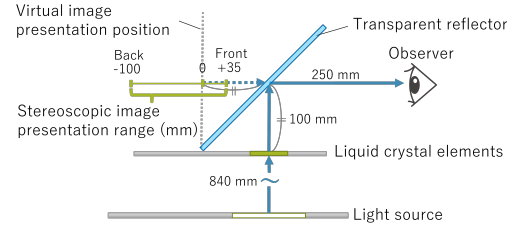


Figure 4: Diagram of prototype display (side view).

A questionnaire was conducted to evaluate the generated stereoscopic images. A three-dimensional model was presented at several depth positions to five subjects. The subjects were asked to record the depth position at which they could see the three-dimensional model. The results are shown in Figure 5(b). As shown, the subjects accurately perceived the depth presentation position of the stereoscopic image. Similar results were obtained at other observation positions.

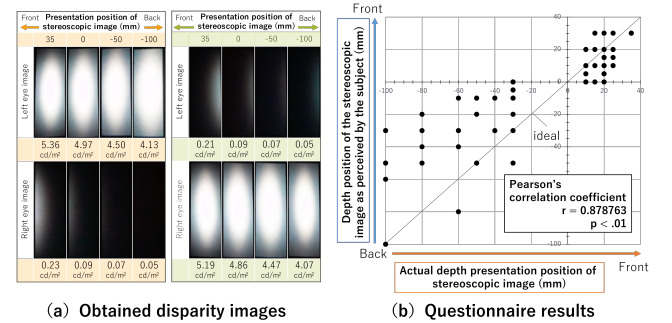


Figure 5: Experimental results.

In addition, we were able to produce a stereoscopic image with a sense of presence when the image was presented in such a way that it blended into the surrounding environment (Figure 1(c)). Compared to a flat image, a stereoscopic image creates a greater sense of depth and blends more naturally into its surroundings.

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